Passive Reconnaissance

three command-line tools:

- `whois` to query WHOIS servers
- `nslookup` to query DNS servers
- `dig` to query DNS servers

`whois` to query WHOIS records

nslookup `and` dig` to query DNS database records

- DNSDumpster
- Shodan.io

Passive VS Active Recon

Passive

passive reconnaissance, you rely on publicly available knowledge

Passive reconnaissance activities include many activities, for instance:

- Looking up DNS records of a domain from a public DNS server.
- Checking job ads related to the target website.
- Reading news articles about the target company.

Active

Active reconnaissance, on the other hand, cannot be achieved so discreetly. It requires direct engagement with the target.

Examples of active reconnaissance activities include:

- Connecting to one of the company servers such as HTTP, FTP, and SMTP.
- Calling the company in an attempt to get information (social engineering).
- Entering company premises pretending to be a repairman

Whois

WHOIS server listens on TCP port 43 for incoming requests. The domain registrar is responsible for maintaining the WHOIS records for the domain names it is leasing. The WHOIS server replies with various information related to the domain requested

- Registrar: Via which registrar was the domain name registered?
- Contact info of registrant: Name, organization, address, phone, among other things. (unless made hidden via a privacy service)
- Creation, update, and expiration dates: When was the domain name first registered? When was it last updated? And when does it need to be renewed?
- Name Server: Which server to ask to resolve the domain name?

The syntax is `whois DOMAIN_NAME`

nslookup and dig

Find the IP address of a domain name using `nslookup`, which stands for Name Server Look Up

`nslookup DOMAIN_NAME`

`nslookup OPTIONS DOMAIN_NAME SERVER`. These three main parameters are:

- OPTIONS contains the query type as shown in the table below. For instance, you can use `A` for IPv4 addresses and `AAAA` for IPv6 addresses.
- DOMAIN_NAME is the domain name you are looking up.
- SERVER is the DNS server that you want to query. You can choose any local or public DNS server to query. Cloudflare offers `1.1.1.1` and `1.0.0.1`, Google offers `8.8.8.8` and `8.8.4.4`, and Quad9 offers `9.9.9.9` and `149.112.112.112`. There are many [more public DNS servers(opens in new tab)]() that you can choose from if you want alternatives to your ISP's DNS servers.

Query type	Result
A	IPv4 Addresses
AAAA	IPv6 Addresses
CNAME	Canonical Name
MX	Mail Servers
SOA	Start of Authority
TXT	TXT Records

nslookup -type=A tryhackme.com 1.1.1.1

Terminal

```
user@TryHackMe$ nslookup -type=A tryhackme.com 1.1.1.1
Server:     1.1.1.1
Address:    1.1.1.1#53

Non-authoritative answer:
Name:   tryhackme.com
Address: 172.67.69.208
Name:   tryhackme.com
Address: 104.26.11.229
Name:   tryhackme.com
Address: 104.26.10.229
```

This lookup is helpful to know from a penetration testing perspectiv

started with one domain name, and we obtained three IPv4 addresses. Each of these IP addresses can be further checked for insecurities

nslookup -type=MX tryhackme.com

```
Terminal

user@TryHackMe$ nslookup -type=MX tryhackme.com
Server:     127.0.0.53
Address:    127.0.0.53#53

Non-authoritative answer:
tryhackme.com   mail exchanger = 5 alt1.aspmx.l.google.com.
tryhackme.com   mail exchanger = 1 aspmx.l.google.com.
tryhackme.com   mail exchanger = 10 alt4.aspmx.l.google.com.
tryhackme.com   mail exchanger = 10 alt3.aspmx.l.google.com.
tryhackme.com   mail exchanger = 5 alt2.aspmx.l.google.com.
```

For more advanced DNS queries and additional functionality, you can use `dig`

`dig DOMAIN_NAME`

but to specify the record type, we would use `dig DOMAIN_NAME TYPE`.

Optionally, we can select the server we want to query using `dig @SERVER DOMAIN_NAME TYPE`

```
Terminal

user@TryHackMe$ dig tryhackme.com MX

; <<>> DiG 9.16.19-RH <<>> tryhackme.com MX
;; global options: +cmd
;; Got answer:
;; ->>HEADER<
```

task

```
C:\Users\Jake>nslookup -type=TXT thmlabs.com
Server:  cache1.service.virginmedia.net
Address:  194.168.4.100

Non-authoritative answer:
thmlabs.com     text =

        "THM{a5b83929888ed36acb0272971e438d78}"
```

i used the nslookup command for a txt file on the thmlab.com domain

DNSDumpster

DNS lookup tools, such as nslookup and dig, cannot find subdomains on their own

[DNSDumpster]

If we search DNSDumpster for `tryhackme.com`, we will discover the subdomain `blog.tryhackme.com`, which a typical DNS query cannot provide. In addition, DNSDumpster will return the collected DNS information in easy-to-read tables and a graph

list of DNS servers for the domain we are looking up. DNSDumpster also resolved the domain names to IP addresses and even tried to geolocate them

MX records; DNSDumpster resolved all five mail exchange servers to their respective IP addresses and provided more information about the owner and location

Finally, we can see TXT records. Practically a single query was enough to retrieve all this information.

Shodan.io

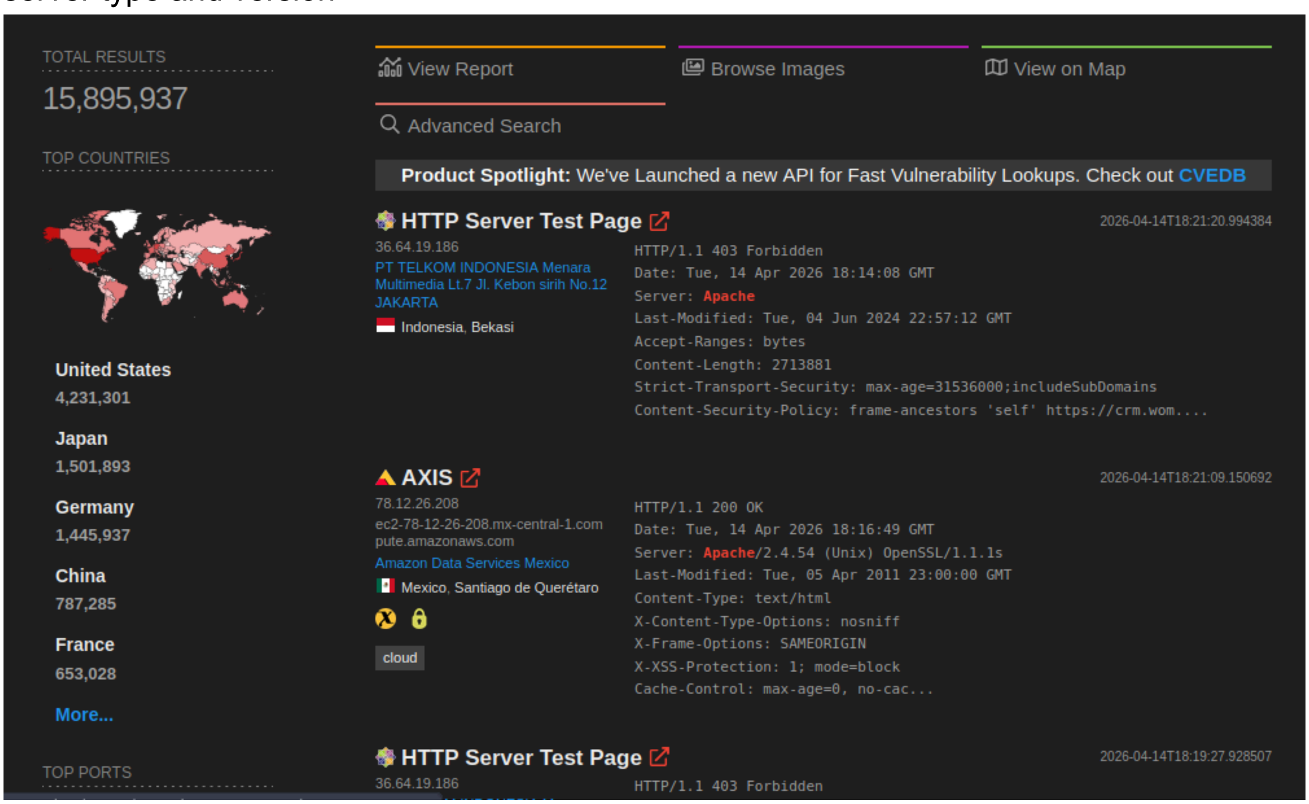
When you are tasked to run a penetration test against specific targets, as part of the passive reconnaissance phase, a service like Shodan.io(opens in new tab) can be helpful to learn various pieces of information about the client's network, without actively connecting to it

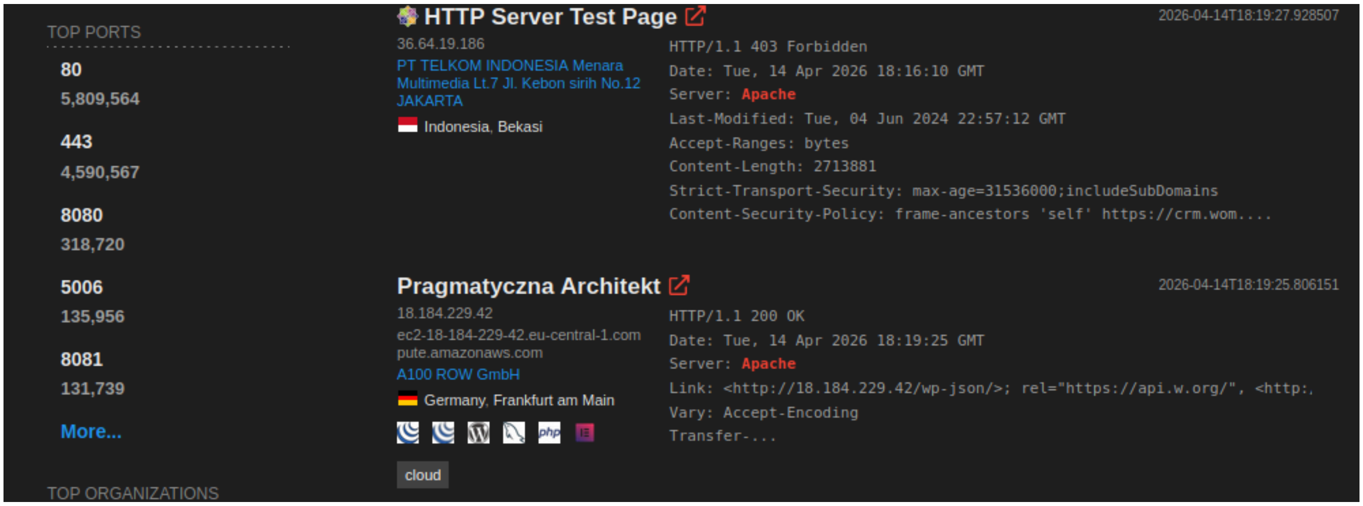
on the defensive side, you can use different services from Shodan.io to learn about connected and exposed devices belonging to your organisation.

Shodan.io tries to connect to every device reachable online to build a search engine of connected "things" in contrast with a search engine for web pages. Once it gets a response, it collects all the information related to the service and saves it in the database to make it searchable

Via this Shodan.io search result, we can learn several things related to our search, such as:

- IP address
- hosting company
- geographic location
- server type and version





Purpose	Commandline Example
Lookup WHOIS record	`whois tryhackme.com`
Lookup DNS A records	`nslookup -type=A tryhackme.com`
Lookup DNS MX records at DNS server	`nslookup -type=MX tryhackme.com 1.1.1.1`
Lookup DNS TXT records	`nslookup -type=TXT tryhackme.com`
Lookup DNS A records	`dig tryhackme.com A`
Lookup DNS MX records at DNS server	`dig @1.1.1.1 tryhackme.com MX`
Lookup DNS TXT records	`dig tryhackme.com TXT`